11-777 Project Preproposal: Efficient Multimodal Reasoning

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1. Project Repository

https://github.com/alvinwang1/multimodal-reasoning-project/

2. Problem Definition (0.5 Point)

2.1. Research Problem

We study *efficient multimodal reasoning*: how a model can exploit several sensory streams while minimizing the training and inference cost of doing so. An embodied agent carrying cameras, microphones, depth sensors, and tactile arrays cannot afford to encode every stream at full resolution on every decision step, yet it also cannot afford to discard a stream that turns out to carry the decisive evidence. The operating question is therefore not whether multimodal input helps, but what the cheapest sufficient encoding of that input is for a given task.

We ground this question in OmniBench (1), a tri-modal benchmark pairing an image, an audio clip, and a text question with four answer options. OmniBench is unusually well suited to a cost study because its annotation protocol guarantees that the correct answer requires information from both the image and the audio; neither alone suffices. This removes the usual confound in efficiency work, where a model that ignores a modality appears efficient only because the benchmark never needed that modality. Here, any saving must be paid for out of genuinely necessary signal, so the accuracy-cost tradeoff is real rather than an artifact of benchmark construction.

2.2. Multimodal

The problem is inherently multimodal rather than a generic efficiency method applied to multimodal data. Every quantity we optimize is defined across modalities: the routing decision is a choice among sensory streams, the adaptation target is a modality the backbone does not yet have, and the cache compression policy is only interesting because different modalities produce token sequences with different lengths, redundancy, and value per token. None of these has a single-modality analogue. We expect the optimal cost policy to differ across these three audio types, which turns efficiency into a question about the structure of cross-modal information rather than a matter of engineering throughput.

2.3. Research Ideas

1. *Lightweight modality routing.* Rather than asking whether a modality is needed, which OmniBench answers affirmatively by construction, we ask *how much* of each modality is needed. We will train a small router that inspects the question and cheap low-resolution previews of each stream, then allocates an encoding budget: full resolution, downsampled, or text-substituted per modality. Evaluation is the compute saved at a fixed accuracy floor, and we will additionally test the binary variant on a mixed pool that combines OmniBench items with single-modality-sufficient items drawn from OmniInstruct, where a genuine skip decision exists.

2. *Sample-efficient modality learning.* Can an agent acquire a new modality with limited data and lightweight adaptation? We will start from an existing vision-language model and add audio through adapters, a learned projector, or LoRA, training only the new parameters on OmniInstruct while the backbone stays frozen. The measurement of interest is the data-efficiency curve: accuracy as a function of adaptation samples and trainable parameter count, which tells us the marginal cost of a modality rather than only its final benefit.

3. *Modality-aware KV-cache optimization.* Multimodal prefills are dominated by vision and audio tokens, so the decoder cache is where inference cost concentrates. We will compare full KV caching against compression policies that allocate retention budget by modality and by attention mass, and measure the resulting accuracy, peak memory, and latency. We expect the tolerable compression ratio to differ sharply between speech, which is linguistically redundant, and sound events, where a short segment may carry the whole answer.

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3. Dataset and Baselines (0.5 Point)

3.1. Selected Dataset

OmniBench contains 1,142 human-annotated multiple-choice items released as a single split, which we have downloaded and verified. Every item has exactly four options. Each record provides an image, an audio clip, a question, four options, one of 8 task type labels, and one of 3 audio type labels (speech, sound event, music). Each record also ships `audio content`, a human transcript, and `image content`, a VLM-generated caption.

We chose this pair because it separates the two things an efficiency study needs. OmniBench supplies a clean evaluation where both modalities are provably necessary, and OmniInstruct supplies roughly 84.5K tri-modal instruction tuning samples, without which Ideas 1 and 2 would have no training signal. The main limitation is evaluation size: 1,142 items in one split means confidence intervals wide enough that small accuracy differences are not meaningful. Since our claims are about cost at matched accuracy rather than about accuracy itself, this is tolerable, but we will report bootstrap confidence intervals throughout and hold out a stratified 20% development split for prompt and hyperparameter selection so that tuning does not contaminate final numbers. We encountered no download issues.

3.2. Data Analysis

Figure 1 reports our inspection of the downloaded data. The left panel gives the joint distribution of the 8 task types against the 3 audio types. Two properties constrain our experimental design. First, audio type is heavily skewed: 771 items use speech, 265 use sound events, and only 106 use music. Since we expect compression tolerance to differ sharply between linguistically redundant speech and information-dense sound events, this imbalance means our per-audio-type results will have very different precision, and the music condition in particular will support only directional claims. Second, the task distribution has a long tail: the five largest types hold 1,070 of the 1,142 items, while Identity and Relationship, Text and Symbols, and Count and Quantity hold 32, 25, and 15 respectively. Several cells are empty outright, with no music items for Identity and Relationship and no sound event items for the two smallest types. We will therefore report efficiency results pooled and broken out by audio type, but not by the full task-by-audio grid.

The right panel is a shortcut audit motivated by our cost claims. If some items were partly answerable from the question text alone, a compression policy could discard real sensory signal and still look efficient. For each item we compute the Jaccard overlap between the question tokens and the correct option, against the mean overlap with the three distractors. The means are 0.0361 and 0.0371 respectively, a difference of -0.001, which is both negligible and in the wrong direction to constitute a shortcut. We take this as evidence that accuracy on this benchmark cannot be obtained from lexical association with the question, which is a precondition for interpreting any saving we report as genuine.

3.3. Baselines

1. *Unified-IO 2* (2), https://arxiv.org/abs/2312.17172. An encoder-decoder trained from scratch over a shared token space spanning image, text, audio, and action. It is the strongest fully open native OLM reported on OmniBench, scoring 33.98% at xxlarge (6.8B). Its near-zero gain from the second modality makes it our primary target: it is already paying full multimodal cost for almost no multimodal benefit.

2. *video-SALMONN* (3), https://arxiv.org/abs/2406.15704. An audio-visual LLM built for video with joint speech, audio event, and visual processing, scoring 35.64% at 13B. It represents the audio-visual-first design lineage and is the natural starting point for Idea 2, since its encoder stack is already organized around adding audio to a visual backbone.

3. *AnyGPT* (4), https://arxiv.org/abs/2402.12226. A discrete-token any-to-any model quantizing each modality into a shared vocabulary consumed by an LLM backbone. At 7B it scores 18.04%, below the chance floor, which makes it a lower anchor and a useful check on whether failures reflect instruction-following breakdown rather than reasoning capacity. Its discrete tokenization also gives Idea 3 a second, structurally different cache regime to compress.

The published Gemini-1.5-Pro result of 47.56% serves as an approximate proprietary upper reference, though we cannot measure its inference cost.

3.4. Compute Requirements

Storage. OmniBench media is a few GB; OmniInstruct with media is substantially larger. We budget roughly 200 GB of scratch and will stream or subsample OmniInstruct if quota becomes binding.

GPU. A 7B OLM in bf16 fits on a single 48 GB GPU; video-SALMONN at 13B needs 80 GB or 4-bit quantization. Inference across three baselines, 1,142 items, and our routing and compression conditions is on the order of 30 GPU-hours. Training is confined to small components: the Idea 1 router is a lightweight head over frozen previews,

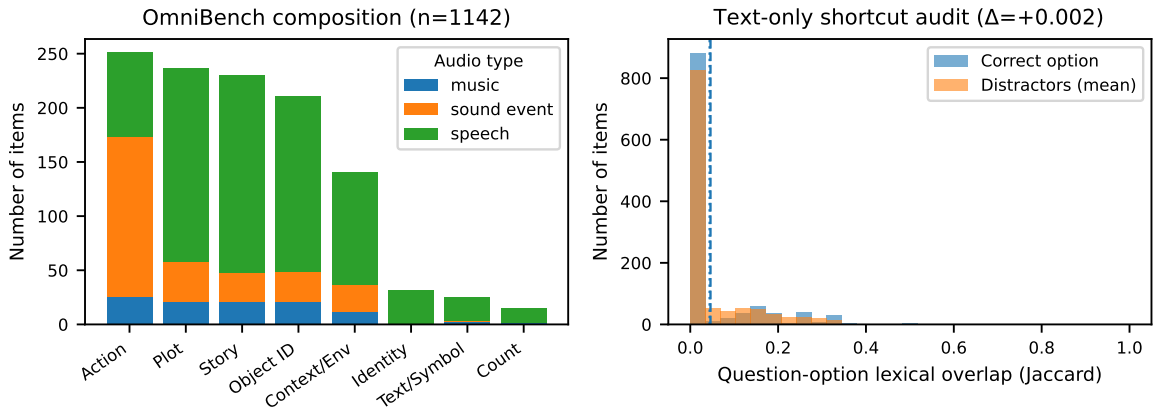


Figure 1. Left: item counts by task type and audio type, showing the speech-dominated audio distribution and the long tail of small task types. Right: question-to-option lexical overlap for correct answers versus distractors. The gap of -0.001 indicates no text-only lexical shortcut.

and Idea 2 trains only adapters or LoRA parameters with the backbone frozen, both of which fit comfortably on one 48 GB GPU.

Measurement protocol. Because our claims are about latency and memory rather than accuracy alone, all timing runs must execute on identical hardware with fixed batch size and no co-tenancy, which we will schedule as exclusive-access jobs. Numbers from quantized or shared runs will be reported separately and marked as such.

Contingency. If GPU access is constrained, we will reduce to two baselines at 7B, subsample OmniInstruct to roughly 20K examples stratified by audio type, and drop the Idea 2 data-efficiency sweep to a small number of points. Ideas 1 and 3 are largely inference-time and degrade gracefully.

References

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